

RETAILMART V3 • SQL

JOINS Part 2 — Quick Quiz

Answer Before Looking!

Test your JOIN intuition. Each question targets a key concept.

Quiz 1: When to Use FULL OUTER JOIN?

EASY

When would you choose FULL OUTER JOIN over LEFT JOIN?

ANSWER: When you need unmatched rows from BOTH sides. LEFT JOIN preserves only the left-side unmatched. FULL OUTER preserves both — essential for reconciliation work where either side could have orphan rows.

Quiz 2: 5-Table JOIN Row Count

MEDIUM

You JOIN orders (1000) → customers (500) → items (5000) → products (2000) → stores (50). All INNER JOINs. Roughly how many rows will the result have?

ANSWER: Driven by the one-to-many relationships. Orders → items is one-to-many (avg 5 items per order), so 1000 orders × 5 items ≈ 5000 rows. Customers, products, stores are one-to-one dimensions — they don't multiply rows. So roughly 5000 result rows, NOT 1000.

Quiz 3: Why Did My LEFT JOIN Lose Rows?

MEDIUM

ANSWER: The LEFT JOIN silently became an INNER JOIN. WHERE o.net_total > 1000 removes all rows where net_total is NULL — including the unmatched customers. Fix: move the filter into the ON clause, or use COALESCE in WHERE.

Question



ANSWER

```
SELECT c.*, o.net_total  
FROM customers.customers c  
LEFT JOIN sales.orders o  
  ON c.customer_id = o.cust_id  
WHERE o.net_total > 1000;
```

Quiz 4: Three JOIN Rewrites

HARD

'Find orders with customer names from Mumbai.' Name three different ways to write this.

ANSWER: (1) JOIN customers and addresses, filter WHERE city = 'Mumbai'. (2) Subquery in WHERE: WHERE cust_id IN (SELECT customer_id FROM addresses WHERE city = 'Mumbai'). (3) Subquery in FROM: JOIN to a pre-filtered (SELECT * FROM customers c JOIN addresses a... WHERE a.city = 'Mumbai') derived table. All return the same result; performance varies.

Quiz 5: Supply Chain Trace Count

CRAZY

Supply chain has 3 warehouses, 5 shipments per warehouse, 100 products in inventory per warehouse. A query joins all three. If you don't filter, how many rows?

ANSWER: $3 \text{ warehouses} \times 5 \text{ shipments} \times 100 \text{ inventory} = 1,500$ rows BEFORE any filtering. But this is a Cartesian-ish situation — without proper FK matching, you can blow up to $3 \times 5 \times 100 \times \dots$ pretty quickly. ALWAYS check intermediate row counts when chaining 4+ tables.